VOLUME I: ADVANCED MATHEMATICAL FOUNDATIONS

CHAPTER 1: Advanced Vector Mathematics, Linear Spaces and Matrix Calculus

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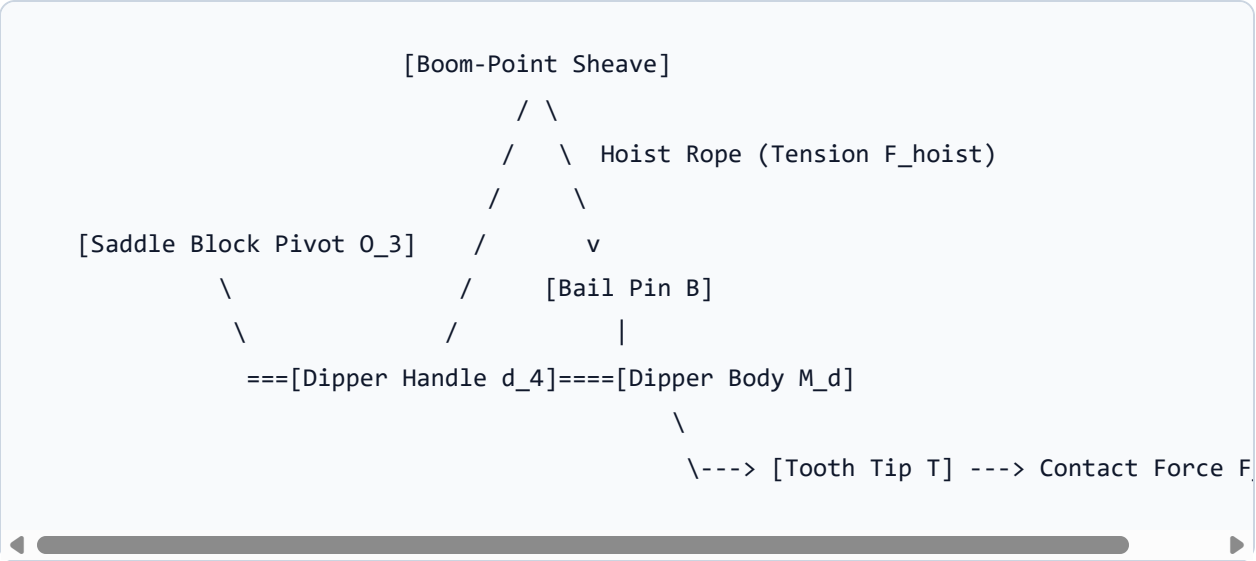
Target Machine Context: Machine A (P&H 2100BL Benchmark) & Machine B (TZ WK-20 Deployment)

1.1 Physical Intuition & Engineering Motivation in Mining Multibody Dynamics

In an electric rope shovel, forces and motions originate at widely separated physical locations:

- High-voltage electrical induction motors apply torque to hoist and crowd drums inside the machinery house.
- Steel wire ropes transmit tension over boom-point sheaves situated more than $16\,\mathrm{m}$ above the ground.
- The dipper handle translates linearly through a pivoting saddle block, sweeping a dipper weighing over $50\,\mathrm{metric\ tonnes}$ through rock, earth, and blasted ore.

- Digging resistance forces act at the dipper teeth and bail pin at continuously varying spatial orientations.



To calculate the structural stress in the boom, determine whether the hoist rope is about to exceed its yield limit, or reconstruct the rock resistance force from motor currents, you cannot rely on isolated scalar equations. The dipper teeth move along curved paths where velocity, acceleration, and contact force have components in multiple directions simultaneously.

Every physical vector (displacement, linear velocity, angular velocity, force, moment) is an invariant geometric entity that exists independently of human coordinate systems. However, to compute on an industrial Programmable Logic Controller (PLC) or in a digital twin simulation, these vectors must be projected onto specific **reference frames**.

If an engineer confuses the reference frame in which a force is expressed with the reference frame in which a displacement is measured, the calculated actuator torques will be completely erroneous. **Advanced vector mathematics and matrix calculus provide the exact, unambiguous language needed to transform physical quantities between moving bodies without geometric distortion.**

1.2 Vector Spaces, Basis Sets, and Coordinate Frame Definitions

1.2.1 Real Vector Spaces $\mathbb{R}^n$

A real vector space $\mathcal{V} = \mathbb{R}^n$ over the field of real numbers $\mathbb{R}$ is a set of elements called vectors, equipped with two operations:

1. **Vector Addition:** $+ : \mathcal{V} \times \mathcal{V} \to \mathcal{V}$, satisfying commutativity, associativity, existence of a zero vector $\mathbf{0}$, and existence of an additive inverse $-\mathbf{v}$.
2. **Scalar Multiplication:** $\cdot : \mathbb{R} \times \mathcal{V} \to \mathcal{V}$, satisfying associativity and distributivity over vector and scalar addition.

In the 3D physical workspace of the shovel, physical positions and forces inhabit $\mathbb{R}^3$. When we consider the joint configuration of the mechanism (e.g., the saddle angle q_3 and crowd extension d_4), the system configuration inhabits an n-dimensional configuration manifold $\mathcal{Q} \subset \mathbb{R}^n$ (for the planar shovel model, $n = 2$).

1.2.2 Basis Vectors and the Right-Handed Cartesian Frame

A set of vectors $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$ in $\mathbb{R}^3$ forms an **orthonormal basis** if and only if:

$$\mathbf{e}_i \cdot \mathbf{e}_j = \delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases} \tag{1.1}$$

where δ_{ij} is the Kronecker delta.

The basis is defined as **right-handed** if the vector cross product satisfies:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \quad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \quad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2 \tag{1.2}$$

Any spatial vector $\mathbf{r} \in \mathbb{R}^3$ is written as a unique linear combination of these basis vectors:

$$\mathbf{r} = x\,\mathbf{e}_1 + y\,\mathbf{e}_2 + z\,\mathbf{e}_3 \tag{1.3}$$

where the scalar projections $x, y, z \in \mathbb{R}$ are the coordinates of $\mathbf{r}$.

1.2.3 Notation for Coordinate Frames

Throughout this research, we establish an explicit notation for reference frames:

- A coordinate frame A consists of an origin O_A and an orthonormal triad of unit vectors $\{\mathbf{x}_A, \mathbf{y}_A, \mathbf{z}_A\}$.
- The representation of a vector $\mathbf{p}$ with respect to Frame A is denoted by the leading superscript ${}^A\mathbf{p}$:

$${}^A\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix}_A = \begin{bmatrix} \mathbf{p} \cdot \mathbf{x}_A \\ \mathbf{p} \cdot \mathbf{y}_A \\ \mathbf{p} \cdot \mathbf{z}_A \end{bmatrix} \in \mathbb{R}^{3\times 1} \tag{1.4}$$

- The vector from point O_A to point O_B, expressed in Frame A, is denoted ${}^A\mathbf{p}_{B/A}$ (or simply ${}^A\mathbf{p}_B$).

1.2.4 The Shovel Reference Frame Hierarchy

To eliminate coordinate ambiguity across milestones M1, M2, and M3, we establish the official project reference frame hierarchy:

```
[Frame 0: World Ground] ---> [Frame 1: Machinery House / Revolving Deck]
                                          |
                                          v
                                [Frame 2: Boom Assembly]
                                          |
                                          v
                                [Frame 3: Saddle Block (Revolute Pivot O_3)]
                                          |
                                          v
                                [Frame 4: Dipper Handle & Bucket (Prismatic Slide)]
```

Frame Designation	Origin Location	Axis Orientation Convention	Physical Attachment
Frame 0 ($\mathcal{F}_0$): World Ground	Fixed point on the mine floor / bench surface.	$+Z_0$ vertically upward opposing gravity; $+X_0$ horizontal toward initial dig face; $+Y_0$ completing right-handed triad.	Static earth datum.
Frame 1 ($\mathcal{F}_1$): Revolving Deck	Center of the swing roller circle on the revolving deck.	$+Z_1$ aligned with the vertical swing rotation axis; $+X_1$ directed along the longitudinal centerline toward the boom foot pin.	Revolving frame (rotates by swing angle q_1).
Frame 2 ($\mathcal{F}_2$): Boom Structure	Boom foot pin connecting boom to revolving deck.	$+X_2$ directed along the boom neutral axis from foot pin to point sheave; $+Z_2$ parallel to the boom-foot revolute axis.	Fixed boom structure (inclined at constant angle $\theta_{\text{boom}} = 45^\circ$).
Frame 3 ($\mathcal{F}_3$): Saddle Block	Center of the shipper shaft / saddle block pivot axis (O_3).	$+X_3$ directed along the longitudinal axis of the dipper handle guide; $+Z_3$ aligned with the saddle block revolute axis (saddle pitch axis).	Saddle block (rotates relative to horizontal by handle pitch angle q_3).

Frame Designation	Origin Location	Axis Orientation Convention	Physical Attachment
Frame 4 ($\mathcal{F}_4$): Dipper & Handle	Reference datum on the dipper handle ($x_{\text{ref}} = 1.50\,\text{m}$ upstream of bail pin).	Axes parallel to Frame 3, but translating along $+X_3$ by crowd displacement coordinate d_4.	Rigid dipper handle body.

1.3 The Special Orthogonal Group $SO(2)$ and $SO(3)$

1.3.1 Direction Cosine Matrices and $SO(3)$

Consider two coordinate frames, Frame A and Frame B, sharing a common origin but oriented differently in space. A vector $\mathbf{v}$ can be expressed in terms of the basis vectors of Frame A ($\{\mathbf{x}_A, \mathbf{y}_A, \mathbf{z}_A\}$) or Frame B ($\{\mathbf{x}_B, \mathbf{y}_B, \mathbf{z}_B\}$):

$$\mathbf{v} = {}^{A}v_x\mathbf{x}_A + {}^{A}v_y\mathbf{y}_A + {}^{A}v_z\mathbf{z}_A = {}^{B}v_x\mathbf{x}_B + {}^{B}v_y\mathbf{y}_B + {}^{B}v_z$$

To find the components in Frame A given the components in Frame B, take the scalar dot product of Eq. (1.5) with $\mathbf{x}_A$:

$$ {}^{A}v_x = \mathbf{v} \cdot \mathbf{x}_A = {}^{B}v_x(\mathbf{x}_B \cdot \mathbf{x}_A) + {}^{B}v_y(\mathbf{y}_B \cdot \mathbf{x}_A) + {}^{B}v_z(\mathbf{z}_B $$

Repeating this projection for $\mathbf{y}_A$ and $\mathbf{z}_A$ yields the matrix transformation:

$$ \begin{bmatrix} {}^{A}v_x \\ {}^{A}v_y \\ {}^{A}v_z \end{bmatrix} = \begin{bmatrix} \mathbf{x}_B \cdot \mathbf{x}_A & \mathbf{y}_B \cdot \mathbf{x}_A & \mathbf{z}_B \cdot \mathbf{x}_A \\ \mathbf{x}_B \cdot \mathbf{y}_A & \mathbf{y}_B \cdot \mathbf{y}_A & \mathbf{z}_B \cdot \mathbf{y}_A \\ \mathbf{x}_B \cdot \mathbf{z}_A & \mathbf{y}_B \cdot \mathbf{z}_A & \mathbf{z}_B \cdot \mathbf{z}_A \end{bmatrix} \begin{bmatrix} {}^{B}v_x \\ {}^{B}v_y \\ {}^{B}v_z \end{bmatrix} \tag{1.7} $$

This 3×3 matrix is the **Rotation Matrix** (or Direction Cosine Matrix) representing the orientation of Frame B relative to Frame A, denoted ${}^{A}\mathbf{R}_B$:

$$ {}^{A}\mathbf{v} = {}^{A}\mathbf{R}_B \, {}^{B}\mathbf{v} \tag{1.8} $$

The columns of ${}^{A}\mathbf{R}_B$ are the unit basis vectors of Frame B projected into Frame A:

$$ {}^{A}\mathbf{R}_B = \begin{bmatrix} {}^{A}\mathbf{x}_B & {}^{A}\mathbf{y}_B & {}^{A}\mathbf{z}_B \end{bmatrix} \in \mathbb{R}^{3\times3} \tag{1.9} $$

The set of all such rotation matrices forms the **Special Orthogonal Group** of order 3, denoted $SO(3)$:

$$SO(3) = \left\{ \mathbf{R} \in \mathbb{R}^{3\times 3} \;\middle|\; \mathbf{R}^T\mathbf{R} = \mathbf{I}_3,\ \det(\mathbf{R}) = +1 \right\} \quad (1$$

1.3.2 Canonical Single-Axis Rotations

The elementary rotation matrices representing a counter-clockwise rotation by an angle θ about the primary coordinate axes are:

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \tag{1.11a}$$

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \tag{1.11b}$$

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.11c}$$

1.3.3 The Planar Shovel Group $SO(2)$

In Milestones M1 and M2, the baseline dynamics of the shovel are modelled in the vertical digging plane (2-DOF saddle pitch and crowd extension). When the motion is confined to the vertical $X - Y$ plane, the rotation occurs strictly about the transverse Z-axis. The rotation matrix reduces to the 2×2 Special Orthogonal Group $SO(2)$:

$$\mathbf{R}(q_3) = \begin{bmatrix} \cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3 \end{bmatrix} \in SO(2) \qquad (1.12)$$

where q_3 is the handle pitch angle measured counter-clockwise from the horizontal forward axis.

1.4 Fundamental Properties of Orthogonal Matrices

Rotation matrices possess mathematical properties that simplify multibody equations of motion:

Property 1: The Orthogonality Condition

Because the columns of $\mathbf{R} = \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix}$ are mutually orthonormal unit vectors ($\mathbf{r}_i \cdot \mathbf{r}_j = \delta_{ij}$):

$$\mathbf{R}^T\mathbf{R} = \begin{bmatrix} \mathbf{r}_1^T \\ \mathbf{r}_2^T \\ \mathbf{r}_3^T \end{bmatrix} \begin{bmatrix} \mathbf{r}_1 & \mathbf{r}_2 & \mathbf{r}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{r}_1 \cdot \mathbf{r}_1 & \mathbf{r}_1 \cdot \mathbf{r}_2 & \mathbf{r}_1 \cdot \mathbf{r}_3 \\ \mathbf{r}_2 \cdot \mathbf{r}_1 & \mathbf{r}_2 \cdot \mathbf{r}_2 & \mathbf{r}_2 \cdot \mathbf{r}_3 \\ \mathbf{r}_3 \cdot \mathbf{r}_1 & \mathbf{r}_3 \cdot \mathbf{r}_2 & \mathbf{r}_3 \cdot \mathbf{r}_3 \end{bmatrix}$$

Property 2: Inversion via Transposition

From Eq. (1.13), the matrix inverse of any rotation matrix is equal to its matrix transpose:

$$\mathbf{R}^{-1} = \mathbf{R}^T \tag{1.14}$$

For transformations between Frame A and Frame B:

$$^B\mathbf{R}_A = \left(^A\mathbf{R}_B\right)^{-1} = \left(^A\mathbf{R}_B\right)^T \tag{1.15}$$

Physical Significance: Inverting a coordinate transformation requires zero floating-point matrix inversions (Gaussian elimination). You transpose the matrix by swapping rows and columns.

Property 3: Preservation of Lengths (Isometry)

The Euclidean norm of a transformed vector is invariant under orthogonal transformation:

$$\|{}^A\mathbf{v}\|_2^2 = ({}^A\mathbf{v})^T({}^A\mathbf{v}) = ({}^A\mathbf{R}_B\,{}^B\mathbf{v})^T({}^A\mathbf{R}_B\,{}^B\mathbf{v}) = ({}^B\mathbf{v})^T$$

Rotating a coordinate system cannot change the physical length of a dipper handle, the magnitude of a digging force, or the kinetic energy of a moving body.

Property 4: Preservation of Inner Products and Angles

For any two spatial vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^3$:

$$ {}^A\mathbf{u} \cdot {}^A\mathbf{v} = ({}^A\mathbf{u})^T({}^A\mathbf{v}) = ({}^B\mathbf{u})^T\,{}^A\mathbf{R}_B^T\,{}^A\mathbf{R}_B\,{}^B\mathbf{v} = {}^B\mathbf{u} \cdot {}^B$$

The physical angle $\theta = \arccos\left(\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|}\right)$ between two structural members is identical regardless of the frame in which components are evaluated.

Property 5: The Determinant Constraint

Taking the determinant of both sides of $\mathbf{R}^T\mathbf{R} = \mathbf{I}$:

$$\det(\mathbf{R}^T\mathbf{R}) = \det(\mathbf{R}^T)\det(\mathbf{R}) = (\det(\mathbf{R}))^2 = \det(\mathbf{I}) =$$

- Matrices with $\det(\mathbf{R}) = +1$ are **proper rotations** ($SO(3)$), preserving the right-handed chirality of space.

- Matrices with $\det(\mathbf{R}) = -1$ represent a rotation combined with a physical **reflection** (a left-handed system). Reflections are physically inadmissible for rigid body kinematics.

1.5 Eigenvalues, Eigenvectors, and Principal Inertia Transformations

1.5.1 The Algebraic Eigenvalue Problem

For any square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, a non-zero vector $\mathbf{v} \in \mathbb{C}^n$ is an **eigenvector** with corresponding **eigenvalue** $\lambda \in \mathbb{C}$ if:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v} \iff (\mathbf{A} - \lambda \mathbf{I})\mathbf{v} = \mathbf{0} \tag{1.19}$$

Non-trivial solutions exist if and only if the characteristic determinant vanishes:

$$p(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}) = 0 \tag{1.20}$$

1.5.2 Eigenstructure of a Rotation Matrix $\mathbf{R} \in SO(3)$

Euler's Rotation Theorem states that every 3D spatial rotation has an invariant axis. Let $\mathbf{R} \in SO(3)$ represent a rotation by angle ϕ about a unit axis $\mathbf{n} = [n_x, n_y, n_z]^T$:

1. The eigenvalues of $\mathbf{R}$ are:

$$\lambda_1 = 1, \quad \lambda_{2,3} = e^{\pm j\phi} = \cos\phi \pm j\sin\phi \tag{1.21}$$

1. The real eigenvector corresponding to $\lambda_1 = 1$ is the unit vector $\mathbf{n}$ along the **axis of rotation**:

$$\mathbf{R}\mathbf{n} = 1 \cdot \mathbf{n} = \mathbf{n} \tag{1.22}$$

Points lying on the rotation axis do not change position under the rotation.

1. The trace of $\mathbf{R}$ (the sum of its diagonal elements) is equal to the sum of its eigenvalues:

$$\mathrm{tr}(\mathbf{R}) = R_{11} + R_{22} + R_{33} = 1 + (\cos\phi + j\sin\phi) + (\mathrm{c}$$

This allows the angle of rotation ϕ to be extracted directly from any numerical rotation matrix:

$$\phi = \arccos\left(\frac{\mathrm{tr}(\mathbf{R}) - 1}{2}\right) \tag{1.24}$$

1.5.3 Principal Axes of the Dipper-Handle Inertia Tensor

The mass distribution of the dipper body and handle about its Center of Mass (COM) is defined by the 3×3 **Inertia Tensor** $\mathbf{I}_{\text{COM}}$:

$$\mathbf{I}_{\text{COM}} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \in \mathbb{R}^{3\times 3} \tag{1.25}$$

Because the mass density $\rho(\mathbf{r}) \geq 0$ is physical, $\mathbf{I}_{\text{COM}}$ is a **real symmetric matrix** ($\mathbf{I}_{\text{COM}}^T = \mathbf{I}_{\text{COM}}$). By the Spectral Theorem of linear algebra:

1. All three eigenvalues $\{\lambda_1, \lambda_2, \lambda_3\} = \{I_1, I_2, I_3\}$ are strictly real numbers.
2. The eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ are mutually orthogonal and form the **Principal Axes of Inertia**.
3. Arranging the eigenvectors into a modal matrix $\mathbf{P} = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \mathbf{v}_3] \in SO(3)$ diagonalizes the inertia tensor:

$$\mathbf{P}^T \mathbf{I}_{\text{COM}} \mathbf{P} = \text{diag}(I_1, I_2, I_3) = \begin{bmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{bmatrix} \tag{1.26}$$

In the principal coordinate frame, all products of inertia (I_{xy}, I_{xz}, I_{yz}) vanish identically.

1.6 Symmetric Matrices, Quadratic Forms, and Positive-Definiteness

1.6.1 Quadratic Forms

Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a symmetric matrix ($\mathbf{A} = \mathbf{A}^T$) and $\mathbf{x} \in \mathbb{R}^n$ be a real non-zero vector. The scalar function:

$$Q(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x} = \sum_{i=1}^{n} \sum_{j=1}^{n} A_{ij} x_i x_j \tag{1.27}$$

is called a **quadratic form**.

1.6.2 Definitions of Definiteness

- $\mathbf{A}$ is **Positive-Definite** ($\mathbf{A} \succ 0$) if:

$$\mathbf{x}^T \mathbf{A} \mathbf{x} > 0 \quad \forall\, \mathbf{x} \neq \mathbf{0} \tag{1.28}$$

- $\mathbf{A}$ is **Positive-Semi-Definite** ($\mathbf{A} \succeq 0$) if:

$$\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0 \quad \forall\, \mathbf{x}$$

1.6.3 Sylvester's Criterion

A symmetric matrix $\mathbf{A}$ is positive-definite if and only if all of its leading principal minors (the determinants of the upper-left $k \times k$ submatrices for $k = 1, \ldots, n$) are strictly positive:

$$\Delta_1 = A_{11} > 0, \quad \Delta_2 = \det \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} > 0, \quad \ldots,$$

Alternatively, $\mathbf{A} \succ 0$ if and only if all of its eigenvalues are strictly positive:

$$\lambda_i(\mathbf{A}) > 0 \quad \forall\, i = 1, \ldots, n \tag{1.30}$$

1.6.4 Physical Significance: The Shovel Mass Matrix $\mathbf{D}(q)$

In Milestone M1, the kinetic energy T of the 2-DOF cable shovel mechanism is represented as a quadratic form in the generalized velocities $\dot{\mathbf{q}} = [\dot{q}_3, \dot{d}_4]^T$:

$$T = \frac{1}{2} \dot{\mathbf{q}}^T \mathbf{D}(q) \dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} \dot{q}_3 & \dot{d}_4 \end{bmatrix} \begin{bmatrix} D_{11}(q) & D_{12}(q) \\ D_{21}(q) & D_{22}(q) \end{bmatrix} \begin{bmatrix} \dot{q}_3 \\ \dot{d}_4 \end{bmatrix}$$

For physical systems:

1. **Kinetic energy cannot be zero unless the machine is at rest:**

$$\dot{\mathbf{q}} \neq \mathbf{0} \implies T > 0$$

Therefore, the generalized mass matrix $\mathbf{D}(q)$ **must be strictly positive-definite** ($\mathbf{D}(q) \succ 0$) across every admissible configuration q.

1. **Invertibility:**

Because all eigenvalues of $\mathbf{D}(q)$ are strictly positive:

$$\det(\mathbf{D}(q)) = \prod_{i=1}^{n} \lambda_i > 0 \implies \mathbf{D}^{-1}(q) \text{ exists everywhe}$$

This ensures that the forward dynamics accelerations can be calculated deterministically:

$$\ddot{\mathbf{q}} = \mathbf{D}^{-1}(q) \left[\boldsymbol{\tau} - \mathbf{C}(q, \dot{\mathbf{q}})\dot{\mathbf{q}} - \mathbf{G}(q) - \mathbf{F}(\dot{\mathbf{q}}) \right] \quad (1.33)$$

1.7 Matrix Calculus: Differentiation of Matrix-Valued Functions

In multibody dynamics, matrices depend continuously on time t through the generalized coordinates $q(t)$. We require matrix calculus to take derivatives analytically without confusing indices.

1.7.1 Derivative of an Inner Product

Let $\mathbf{u}(t), \mathbf{v}(t) \in \mathbb{R}^n$ be differentiable vector functions. The derivative of their inner product follows the standard product rule:

$$\frac{d}{dt}\left(\mathbf{u}^T\mathbf{v}\right) = \dot{\mathbf{u}}^T\mathbf{v} + \mathbf{u}^T\dot{\mathbf{v}} \tag{1.34}$$

1.7.2 Derivative of a Quadratic Form

Let $\mathbf{A}(t) \in \mathbb{R}^{n\times n}$ be a symmetric, time-varying matrix and $\mathbf{x}(t) \in \mathbb{R}^n$. The total time derivative of the scalar $Q(t) = \mathbf{x}^T\mathbf{A}(t)\mathbf{x}$ is:

$$\frac{d}{dt}\left(\mathbf{x}^T\mathbf{A}(t)\mathbf{x}\right) = \dot{\mathbf{x}}^T\mathbf{A}\mathbf{x} + \mathbf{x}^T\dot{\mathbf{A}}\mathbf{x} + \mathbf{x}^T\mathbf{A}\dot{\mathbf{x}} = 2\mathbf{x}^T\mathbf{A}\dot{\mathbf{x}} -$$

1.7.3 Gradient with Respect to a Vector

Let $f(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ be a scalar field. The gradient $\nabla_{\mathbf{x}} f$ is defined as the column vector of partial derivatives:

$$\nabla_{\mathbf{x}} f = \frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} \in \mathbb{R}^{n \times 1} \tag{1.36}$$

For linear and quadratic vector products with constant symmetric $\mathbf{A}$:

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{c}^T \mathbf{x}) = \mathbf{c} \tag{1.37}$$

$$\frac{\partial}{\partial \mathbf{x}} (\mathbf{x}^T \mathbf{A} \mathbf{x}) = 2\mathbf{A}\mathbf{x} \tag{1.38}$$

1.7.4 Derivative of a Matrix Inverse

Let $\mathbf{A}(t)$ be an invertible matrix function of time. Differentiating the identity $\mathbf{A}(t)\mathbf{A}^{-1}(t) = \mathbf{I}$:

$$\frac{d}{dt}\left(\mathbf{A}\mathbf{A}^{-1}\right) = \dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\frac{d}{dt}\left(\mathbf{A}^{-1}\right) = \mathbf{0} \tag{1.39}$$

Premultiplying by $\mathbf{A}^{-1}$:

$$\mathbf{A}^{-1}\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}^{-1}\mathbf{A}\frac{d}{dt}\left(\mathbf{A}^{-1}\right) = \mathbf{0} \implies \frac{d}{dt}\left(\mathbf{A}^{-1}\right) =$$

This identity is used when designing momentum observers and Kalman filters that update covariance matrices in real time.

1.8 Time Derivatives of Rotation Matrices and the Skew-Symmetric Operator

1.8.1 Derivation of the Angular Velocity Matrix

Let $\mathbf{R}(t) \in SO(3)$ represent the orientation of a moving body relative to a static frame. By the orthogonality condition (Eq. (1.13)):

$$\mathbf{R}(t)\mathbf{R}^T(t) = \mathbf{I} \tag{1.41}$$

Differentiating with respect to time t:

$$\frac{d}{dt}\left(\mathbf{R}\mathbf{R}^T\right) = \dot{\mathbf{R}}\mathbf{R}^T + \mathbf{R}\dot{\mathbf{R}}^T = \mathbf{0} \implies \dot{\mathbf{R}}\mathbf{R}^T = -\left(\dot{\mathbf{R}}\mathbf{R}\right.$$

Define the matrix $\mathbf{\Omega}(t) \in \mathbb{R}^{3\times 3}$ as:

$$\boldsymbol{\Omega}(t) \triangleq \dot{\mathbf{R}}\mathbf{R}^T \tag{1.43}$$

From Eq. (1.42), $\boldsymbol{\Omega}^T = -\boldsymbol{\Omega}$. Therefore, $\boldsymbol{\Omega}(t)$ is a **skew-symmetric matrix** ($\boldsymbol{\Omega} \in \mathfrak{so}(3)$, the Lie algebra of $SO(3)$).

Postmultiplying Eq. (1.43) by $\mathbf{R}$:

$$\dot{\mathbf{R}}(t) = \boldsymbol{\Omega}(t)\mathbf{R}(t) \tag{1.44}$$

1.8.2 The Skew-Symmetric Vector Operator $S(\boldsymbol{\omega})$ or $[\boldsymbol{\omega}\times]$

Every 3×3 skew-symmetric matrix has exactly three independent parameters and can be mapped to a 3×1 vector $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$:

$$\mathbf{S}(\boldsymbol{\omega}) = [\boldsymbol{\omega}\times] \triangleq \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3} \tag{1.45}$$

The skew-symmetric matrix implements the vector cross product through matrix multiplication:

$$\mathbf{S}(\boldsymbol{\omega})\mathbf{v} = \boldsymbol{\omega} \times \mathbf{v} \quad \forall\, \mathbf{v} \in \mathbb{R}^3 \tag{1.46}$$

Thus, the fundamental kinematic relation governing moving reference frames is:

$$\dot{\mathbf{R}}(t) = \mathbf{S}(\boldsymbol{\omega})\mathbf{R}(t) = [\boldsymbol{\omega}\times]\mathbf{R}(t) \tag{1.47}$$

where $\boldsymbol{\omega}$ is the physical angular velocity vector of the moving body expressed in the reference frame.

1.8.3 Planar Reduction for Shovel Saddle Pitch

For the planar shovel model rotating about the Z-axis with handle pitch rate $\dot{q}_3$:

$$\boldsymbol{\omega} = \begin{bmatrix} 0 \\ 0 \\ \dot{q}_3 \end{bmatrix} \implies \mathbf{S}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\dot{q}_3 & 0 \\ \dot{q}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{1.48}$$

Let us verify Eq. (1.47) directly using 2D rotation matrices:

$$\mathbf{R}(q_3) = \begin{bmatrix} \cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3 \end{bmatrix} \tag{1.49}$$

Differentiating entry by entry using the chain rule:

$$\dot{\mathbf{R}}(q_3) = \frac{d}{dt}\begin{bmatrix}\cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3\end{bmatrix} = \begin{bmatrix}-\sin q_3 \cdot \dot{q}_3 & -\cos$$

Now compute the matrix product $\mathbf{S}(\dot{q}_3)\mathbf{R}(q_3)$:

$$\mathbf{S}(\dot{q}_3)\mathbf{R}(q_3) = \begin{bmatrix}0 & -\dot{q}_3 \\ \dot{q}_3 & 0\end{bmatrix}\begin{bmatrix}\cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3\end{bmatrix} = \begin{bmatrix}-\dot{q}_3 \, \si \\ \dot{q}_3 \, \cos$$

Eq. (1.50) and Eq. (1.51) are identical, proving that the time derivative of a planar rotation matrix is computed by multiplying by the skew-symmetric matrix $\begin{bmatrix}0 & -1 \\ 1 & 0\end{bmatrix}\dot{q}_3$.

1.9 Simple Numerical Worked Example

Problem Statement

A point on a rigid body is located at ${}^{B}\mathbf{p} = [3.0, 4.0]^T \, \mathrm{m}$ in its local body-fixed Frame B. Frame B is rotated by an angle $\theta = 30^\circ$ $(0.5236 \, \mathrm{rad})$ relative to a static reference Frame A.

1. Compute the rotation matrix ${}^{A}\mathbf{R}_{B}$.
2. Calculate the position of the point expressed in Frame A (${}^{A}\mathbf{p}$).
3. Verify that the Euclidean norm is preserved ($\|{}^{A}\mathbf{p}\| = \|{}^{B}\mathbf{p}\|$).

4. Verify the inverse property using the transpose: ${}^{B}\mathbf{p} = ({}^{A}\mathbf{R}_{B})^{T}\,{}^{A}\mathbf{p}$.
5. If the body rotates with an angular rate $\dot{\theta} = 2.0\,\mathrm{rad/s}$, compute the rate of change of the rotation matrix $\dot{\mathbf{R}}$.

Complete Analytical Solution

Step 1: Compute ${}^{A}\mathbf{R}_{B}$

$$\cos(30^\circ) = \frac{\sqrt{3}}{2} \approx 0.866025, \quad \sin(30^\circ) = \frac{1}{2} = 0.500000$$

$$ {}^{A}\mathbf{R}_{B} = \begin{bmatrix} \cos(30^\circ) & -\sin(30^\circ) \\ \sin(30^\circ) & \cos(30^\circ) \end{bmatrix} = \begin{bmatrix} 0.866025 & -0.500000 \\ 0.500000 & 0.866025 \end{bmatrix} $$

Step 2: Transform ${}^{B}\mathbf{p}$ to ${}^{A}\mathbf{p}$

$$ {}^{A}\mathbf{p} = {}^{A}\mathbf{R}_{B}\,{}^{B}\mathbf{p} = \begin{bmatrix} 0.866025 & -0.500000 \\ 0.500000 & 0.866025 \end{bmatrix} \begin{bmatrix} 3.0 \\ 4.0 \end{bmatrix} = \begin{bmatrix} 3.0(0.866 \\ 3.0(0.500 \end{bmatrix} $$

Step 3: Verify Norm Invariance

$$\|{}^B\mathbf{p}\|_2 = \sqrt{3.0^2 + 4.0^2} = \sqrt{9 + 16} = \sqrt{25} = 5.000000\,\text{m}$$

$$\|{}^A\mathbf{p}\|_2 = \sqrt{(0.598076)^2 + (4.964102)^2} = \sqrt{0.357695 + 24.642}$$

The norm is preserved exactly: $\|{}^A\mathbf{p}\|_2 = \|{}^B\mathbf{p}\|_2 = 5.0\,\text{m}$.

Step 4: Verify Inverse Mapping via Transposition

$${}^B\mathbf{p}_{\text{recovered}} = ({}^A\mathbf{R}_B)^T\,{}^A\mathbf{p} = \begin{bmatrix} 0.866025 & 0.500000 \\ -0.500000 & 0.866025 \end{bmatrix} \begin{bmatrix} 0.59807 \\ 4.96410 \end{bmatrix}$$

$${}^Bp_x = 0.866025(0.598076) + 0.500000(4.964102) = 0.517949$$

$${}^Bp_y = -0.500000(0.598076) + 0.866025(4.964102) = -0.299$$

The original vector ${}^B\mathbf{p} = [3.0, 4.0]^T\,\text{m}$ is recovered.

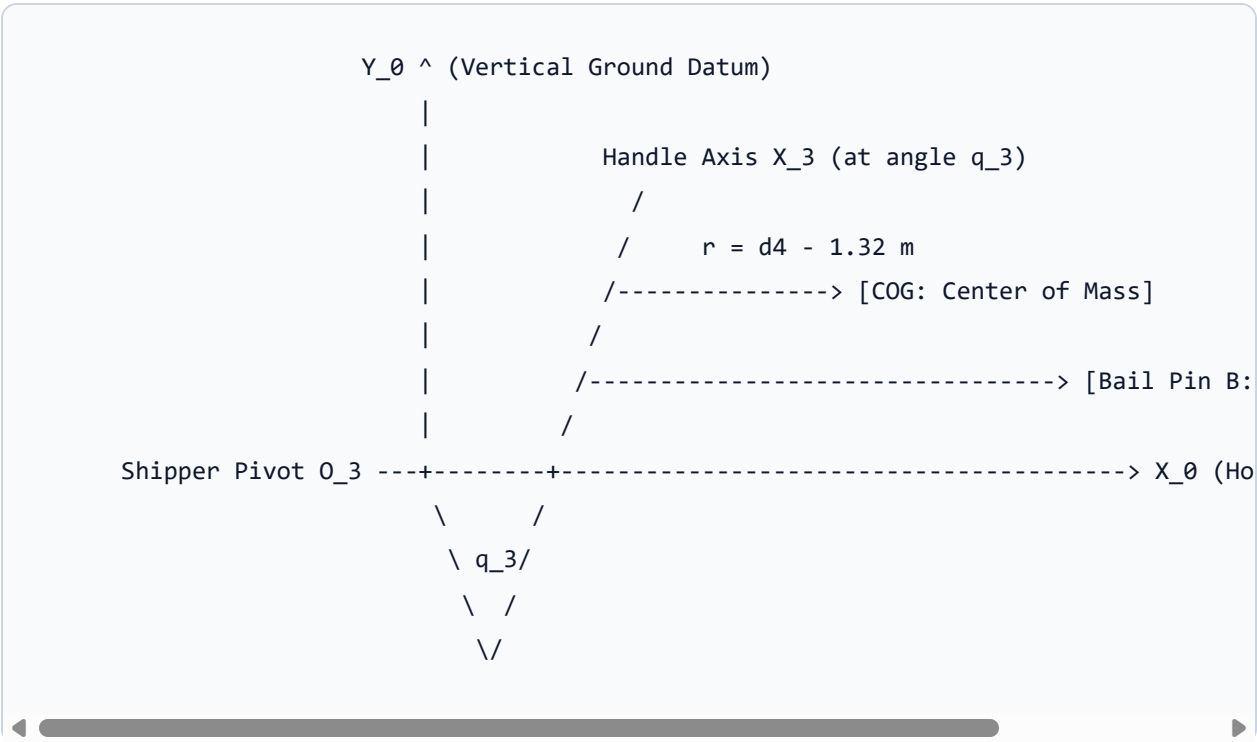
Step 5: Compute $\dot{\mathbf{R}}$ for $\dot{\theta} = 2.0\,\text{rad/s}$

$$\dot{\mathbf{R}} = \mathbf{S}(\dot{\theta})\mathbf{R} = \begin{bmatrix} 0 & -2.0 \\ 2.0 & 0 \end{bmatrix} \begin{bmatrix} 0.866025 & -0.500000 \\ 0.500000 & 0.866025 \end{bmatrix} = \begin{bmatrix} -2.0 \\ 2.0(\end{bmatrix}$$

1.10 Shovel Case Study: Spatial Transformation from Saddle Pivot (O_3) to Bail Pin (B)

Physical Description and Geometric Setup

We now apply these mathematical tools to the benchmark P&H 2100BL shovel model (Milestones M1/M2).



- **Reference Frame $\mathcal{F}_0$ (Origin O_3):**

Located at the saddle block shipper shaft center.

$+X_0$ is horizontal forward toward the digging face; $+Y_0$ is vertical upward against gravity.

- **Reference Frame $\mathcal{F}_3$ (Moving with Handle Pitch):**

Rotates about O_3 by handle angle $q_3(t)$.

$+X_3$ is directed longitudinally along the handle toward the bucket; $+Y_3$ is normal to the handle pointing upward.

- **Physical Offsets (Source: `shovel_params.m`, `claims.md`):**

The crowd coordinate $d_4(t)$ is defined as the prismatic displacement along $+X_3$.

1. The reference datum $x_{\text{ref}} = 1.50\,\text{m}$ defines the bail pin location along $+X_3$:

$$s_{\text{bail}}(t) = d_4(t) + 1.50\,\text{m} \tag{1.52}$$

1. The empty dipper body center of mass (COM) has an offset $L_{\text{COG}} = 2.82\,\text{m}$ upstream from the bail pin. Therefore, the net COG lever arm along $+X_3$ is:

$$r_{\text{COG}}(t) = d_4(t) + c = d_4(t) + (x_{\text{ref}} - L_{\text{COG}}) = d_4(t) -$$

1. The bail pin and the center of mass lie on the handle axis (zero lateral offset in the nominal 2-DOF planar model: $y_3 = 0$).

Mathematical Derivation of Position and Velocity Vectors

Step 1: Define Position Vectors in Local Frame $\mathcal{F}_3$

$$ {}^{3}\mathbf{p}_{\text{bail}}(t) = \begin{bmatrix} d_4(t) + 1.50 \\ 0 \end{bmatrix} \in \mathbb{R}^{2\times 1} \tag{1.54} $$

$$ {}^{3}\mathbf{p}_{\text{COG}}(t) = \begin{bmatrix} d_4(t) - 1.32 \\ 0 \end{bmatrix} \in \mathbb{R}^{2\times 1} \tag{1.55} $$

Step 2: Transform to Fixed Frame $\mathcal{F}_0$

The rotation matrix from Frame 3 to Frame 0 is:

$$ {}^{0}\mathbf{R}_3(q_3) = \begin{bmatrix} \cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3 \end{bmatrix} \tag{1.56} $$

Applying the rotation matrix:

$$ {}^{0}\mathbf{p}_{\text{bail}}(t) = {}^{0}\mathbf{R}_3(q_3)\,{}^{3}\mathbf{p}_{\text{bail}}(t) = \begin{bmatrix} \cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3 \end{bmatrix} \begin{bmatrix} d_4 - \end{bmatrix} $$

$$ {}^0\mathbf{p}_{\text{COG}}(t) = {}^0\mathbf{R}_3(q_3)\,{}^3\mathbf{p}_{\text{COG}}(t) = \begin{bmatrix} \cos q_3 & -\sin q_3 \\ \sin q_3 & \cos q_3 \end{bmatrix} \begin{bmatrix} d $$

Step 3: Differentiate to Obtain Velocity in Fixed Frame $\mathcal{F}_0$

Using the matrix product rule:

$$ {}^0\mathbf{v}_{\text{bail}} = \frac{d}{dt}\left({}^0\mathbf{p}_{\text{bail}}\right) = \frac{d}{dt}\left({}^0\mathbf{R}_3(q_3)\,{}^3\mathbf{p}_{\text{bail}}\right) = \dot{\mathbf{R}}_3\,{}^3\mathbf{p}_{\text{bail}} $$

From Eq. (1.51), $\dot{\mathbf{R}}_3 = \mathbf{S}(\dot{q}_3)\,{}^0\mathbf{R}_3$. Furthermore, ${}^3\dot{\mathbf{p}}_{\text{bail}} = [\dot{d}_4, 0]^T$. Expanding line by line:

$$ {}^0\mathbf{v}_{\text{bail}} = \dot{q}_3 \begin{bmatrix} -\sin q_3 & -\cos q_3 \\ \cos q_3 & -\sin q_3 \end{bmatrix} \begin{bmatrix} d_4 + 1.50 \\ 0 \end{bmatrix} + \begin{bmatrix} \cos q_3 & -\sin q \\ \sin q_3 & \cos q_3 \end{bmatrix} $$

$$ {}^0\mathbf{v}_{\text{bail}} = \begin{bmatrix} -(d_4 + 1.50)\dot{q}_3 \sin q_3 \\ (d_4 + 1.50)\dot{q}_3 \cos q_3 \end{bmatrix} + \begin{bmatrix} \dot{d}_4 \cos q_3 \\ \dot{d}_4 \sin q_3 \end{bmatrix} = \begin{bmatrix} \cos \\ \sin \end{bmatrix} $$

Numerical Evaluation for a Realistic Mining Operating State

Let the shovel be in the mid-cut digging pass with state values:

- Handle pitch: $q_3 = 30^\circ = 0.523599\,\mathrm{rad}$
- Handle angular rate: $\dot{q}_3 = 0.20\,\mathrm{rad/s}$
- Crowd extension: $d_4 = 8.75\,\mathrm{m}$ (nominal baseline bias)
- Crowd extension rate: $\dot{d}_4 = 0.40\,\mathrm{m/s}$

Compute the bail position and velocity in world frame $\mathcal{F}_0$:

1. Spatial Lever Arm:

$$s_{\text{bail}} = 8.75 + 1.50 = 10.25\,\mathrm{m}$$

1. Bail Pin Position in $\mathcal{F}_0$:

$${}^0\mathbf{p}_{\text{bail}} = \begin{bmatrix} 10.25\cos(30^\circ) \\ 10.25\sin(30^\circ) \end{bmatrix} = \begin{bmatrix} 10.25(0.866025) \\ 10.25(0.500000) \end{bmatrix} = \begin{bmatrix} 8.876760 \\ 5.125000 \end{bmatrix}$$

1. Velocity Components in $\mathcal{F}_0$:

$${}^0v_{\text{bail},x} = \dot{d}_4\cos q_3 - s_{\text{bail}}\dot{q}_3\sin q_3 = 0.40(0.866025) - 10.25(0.$$

$$^{0}v_{\text{bail},y} = \dot{d}_4 \sin q_3 + s_{\text{bail}} \dot{q}_3 \cos q_3 = 0.40(0.500000) + 10.25(0.$$

Total linear velocity magnitude:

$$\|^{0}\mathbf{v}_{\text{bail}}\|_2 = \sqrt{(-0.678590)^2 + (1.975352)^2} = \sqrt{0.460484 + 3.9}$$

Physical Interpretation: Even though the crowd drive is actively extending forward at $+0.40\,\text{m/s}$ along the handle, the horizontal velocity in world coordinates is negative ($-0.679\,\text{m/s}$) because the angular pitching motion of the $10.25\,\text{m}$ handle carries the dipper upward and backward relative to the global horizontal axis. Matrix transformations capture this cross-coupling automatically.

1.11 Assumptions, Limitations, and Common Traps

Pitfall / Concept	Common Engineering Mistake	Rigorous Mathematical Reality
Active vs. Passive Rotations	Assuming $\mathbf{R}(\theta)$ rotates a physical vector in space versus transforming the coordinates of a fixed vector.	**Passive rotation (Coordin Transformation):** ${}^{A}\mathbf{p} = {}^{A}\mathbf{R}_{B}\,{}^{B}\mathbf{p}$. **Active rotation (Physical Rotation of an object):** $\mathbf{p}_{\text{new}} = \mathbf{R}\,\mathbf{p}_{\text{old}}$. They ar matrix transposes of each other ($\mathbf{R}_{\text{passive}} = \mathbf{R}^{T}_{\text{acti}}$ Mixing them flips the sign every off-diagonal term.
Non-Commutativity of 3D Rotations	Assuming that 3D rotations commute like 2D rotations: $\mathbf{R}_x(\theta_1)\mathbf{R}_y(\theta_2) \overset{?}{=} \mathbf{R}_y(\theta_2)\mathbf{R}_x(\theta_1)$.	In 3D space, $SO(3)$ is a **n Abelian Lie group**. Matrix multiplication does **not** commute: $\mathbf{R}_A\mathbf{R}_B \neq \mathbf{R}_B\mathbf{R}_A$. The sequence of swing, pitch, a roll operations must follow strictly ordered convention
Unnormalized Vectors in Quadratic Forms	Computing condition numbers or SVD singular values on sensitivity matrices without unit-scaling columns.	In Milestone M2, the sensitivity matrix pairs a torque sensitivity (units $\mathrm{N}\cdot\mathrm{m}/\mathrm{kg} = \mathrm{m}^2/\mathrm{s}^2$) w a load sensitivity (units m) Condition numbers compu without column L_2-normalization depend on whether mass is expressed kg or tonnes. **Columns must be scaled to unit nc**

Pitfall / Concept	Common Engineering Mistake	Rigorous Mathematical Reality
Differentiating Vectors in Rotating Frames	Calculating acceleration by differentiating components twice: $\mathbf{a} \overset{?}{=} [\ddot{x}, \ddot{y}, \ddot{z}]^T$.	When components are expressed in a rotating fra $\mathcal{F}_B$, differentiation require the **Transport Theorem**: $\frac{d}{dt}\mathbf{v} = {}^B\dot{\mathbf{v}} + \boldsymbol{\omega}_{B/A} \times$. Omitting the cross produ drops the Coriolis and centrifugal accelerations.

1.12 Comparison of Related Mathematical Concepts

1.12.1 Rotation Matrix vs. Euler Angles vs. Unit Quaternions

In multibody dynamic modeling of heavy excavators, orientation can be parameterized in three primary mathematical forms:

```
[Orientation Parameterizations]
   ├── 1. Rotation Matrix R in SO(3) [9 parameters, 6 constraints] -> Zero singularit
   ├── 2. Euler Angles (e.g., Z-Y-X) [3 parameters, 0 constraints] -> Suffers from Gi
   └── 3. Unit Quaternions q in S^3  [4 parameters, 1 constraint]  -> Zero singularit
```

- **Rotation Matrices ($\mathbb{R}^{3\times3}$):** Used throughout Milestones M1, M2, and M3. They involve 9 parameters subject to 6 orthonormality constraints ($\mathbf{R}^T\mathbf{R} = \mathbf{I}$). They contain **zero mathematical singularities** and map vectors via standard matrix multiplication.

- **Euler Angles ($\mathbb{R}^3$): **Minimal 3-parameter representation. However, by the hairy ball theorem of differential topology, every 3-parameter orientation system suffers from** Gimbal Lock** (a kinematic singularity where the transformation matrix drops rank and angular rates become indeterminate).
- **Unit Quaternions ($\mathbb{H}$ or S^3):** 4-parameter representation ($\eta \in \mathbb{R}, \boldsymbol{\epsilon} \in \mathbb{R}^3$ with $\eta^2 + \|\boldsymbol{\epsilon}\|^2 = 1$). Used in full 3D real-time gaming engines and flight dynamics. For the planar 2-DOF digging model in M1–M3, planar $SO(2)$ rotation matrices are mathematically superior and singularity-free.

1.13 Practice Problem Set

Problem 1.1 (Foundational: Orthonormality and Frame Inversion)

A coordinate transformation matrix between revolving deck Frame 1 and boom Frame 2 is given by:

$$ {}^1\mathbf{R}_2 = \begin{bmatrix} \cos(45^\circ) & -\sin(45^\circ) & 0 \\ \sin(45^\circ) & \cos(45^\circ) & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

1. Prove analytically that ${}^1\mathbf{R}_2 \in SO(3)$ by verifying that $({}^1\mathbf{R}_2)^T({}^1\mathbf{R}_2) = \mathbf{I}_3$ and $\det({}^1\mathbf{R}_2) = +1$.

2. Given a boom point sheave center located at ${}^{2}\mathbf{p} = [15.50, 0, 0]^{T}\ \mathrm{m}$ in Frame 2, compute the position vector ${}^{1}\mathbf{p}$ expressed in Frame 1.
3. Compute the inverse transformation matrix ${}^{2}\mathbf{R}_{1}$ and demonstrate that ${}^{2}\mathbf{R}_{1}\ {}^{1}\mathbf{p} = {}^{2}\mathbf{p}$.

Problem 1.2 (Intermediate: Eigenstructure and Trace Analysis)

A spatial structural vibration test on the boom-foot pin reveals a rotation matrix between the unloaded and loaded states:

$$\mathbf{R} = \begin{bmatrix} 0.999848 & -0.017452 & 0 \\ 0.017452 & 0.999848 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

1. Compute the trace $\mathrm{tr}(\mathbf{R})$.
2. Using the trace formula (Eq. (1.24)), determine the equivalent structural deflection angle ϕ in degrees.
3. Find the three eigenvalues of $\mathbf{R}$ and verify that the eigenvector corresponding to $\lambda = 1$ is the Z-axis.

Problem 1.3 (Advanced: Quadratic Forms and Inertia Positive-Definiteness)

The generalized mass matrix of a simplified 2-DOF shovel mechanism is given by:

$$\mathbf{D}(d_4) = \begin{bmatrix} I_{zz} + M_d(d_4 + c)^2 & 0 \\ 0 & M_d \end{bmatrix}$$

where dipper mass $M_d = 54,300\,\text{kg}$, moment of inertia $I_{zz} = 287,900\,\text{kg}\cdot\text{m}^2$, and lever offset $c = -1.32\,\text{m}$.

1. Using Sylvester's criterion, prove that $\mathbf{D}(d_4)$ is strictly positive-definite ($\mathbf{D} \succ 0$) for all real values of $d_4 \in (-\infty, \infty)$.
2. Calculate the eigenvalues of $\mathbf{D}(d_4)$ at the fully retracted stroke position ($d_4 = 6.75\,\text{m}$) and the fully extended stroke position ($d_4 = 10.75\,\text{m}$).
3. Compute the ratio of maximum to minimum inertia $\frac{\lambda_{\max}(10.75\,\text{m})}{\lambda_{\max}(6.75\,\text{m})}$ and discuss its significance for joint controller tuning.

Problem 1.4 (Advanced: Skew-Symmetric Operator and Acceleration Propagation)

A sensor measuring point accelerations is mounted on the dipper handle at a fixed position ${}^3\mathbf{r} = [s, 0, 0]^T$ relative to the saddle pivot. The saddle block rotates with time-varying angle $q_3(t)$, angular velocity $\dot{q}_3(t)$, and angular acceleration $\ddot{q}_3(t)$.

1. Derive the expression for the total spatial acceleration vector ${}^0\mathbf{a}$ in the fixed world frame using matrix calculus and the skew-symmetric operator $\mathbf{S}(\boldsymbol{\omega})$.
2. Separate the resulting acceleration terms into tangential acceleration, centripetal acceleration, and explain which term would be measured by a radial accelerometer aligned with $+X_3$.

Problem 1.5 (Research-Grade: The Derivative of a Parameterized Metric Tensor)

In Milestone M3, the generalized momentum is defined as $\mathbf{p} = \mathbf{D}(q)\dot{\mathbf{q}}$. To prove observer passivity, one must compute the matrix difference $\mathbf{N}(q, \dot{q}) = \dot{\mathbf{D}}(q) - 2\mathbf{C}(q, \dot{q})$.

Given the configuration-dependent term $D_{11}(d_4) = I_{zz} + M_d(d_4 + c)^2$:

1. Compute the exact total time derivative $\frac{d}{dt}\left[D_{11}(d_4)\right]$ as a function of the crowd speed $\dot{d}_4$.
2. Given that the corresponding Coriolis matrix entry is $C_{11} = M_d(d_4 + c)\dot{d}_4$, prove that the scalar quantity $\dot{D}_{11} - 2C_{11} \equiv 0$ vanishes identically.
3. Explain why this mathematical cancellation guarantees that the digital twin simulation does not generate artificial numerical energy.

1.14 Complete Analytical Solutions to Practice Problems

Solution 1.1

Part 1: Proof that ${}^1\mathbf{R}_2 \in SO(3)$

Let $\theta = 45^\circ$, $\cos(45^\circ) = \sin(45^\circ) = \frac{1}{\sqrt{2}}$.

$$({}^1\mathbf{R}_2)^T({}^1\mathbf{R}_2) = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Evaluating entry by entry:

- $(1,1): \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 0 = \frac{1}{2} + \frac{1}{2} = 1$
- $(1,2): \left(\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) + 0 = -\frac{1}{2} + \frac{1}{2} = 0$
- $(2,2): \left(-\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + 0 = \frac{1}{2} + \frac{1}{2} = 1$
- $(3,3): 0 + 0 + 1 = 1$

All other off-diagonal terms are zero. Thus $({}^1\mathbf{R}_2)^T({}^1\mathbf{R}_2) = \mathbf{I}_3$.

Now compute the determinant expanding along the third row:

$$\det({}^1\mathbf{R}_2) = 1 \cdot \det \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} = \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right.$$

Since $\mathbf{R}^T\mathbf{R} = \mathbf{I}$ and $\det(\mathbf{R}) = +1$, ${}^1\mathbf{R}_2 \in SO(3)$. ■

Part 2: Position ${}^1\mathbf{p}$

$$
{}^1\mathbf{p} = {}^1\mathbf{R}_2\,{}^2\mathbf{p} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 15.50 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 15.50/\sqrt{2} \\ 15.50/\sqrt{2} \\ 0 \end{bmatrix} \approx
$$

Part 3: Inverse Transformation

$$
{}^2\mathbf{R}_1 = ({}^1\mathbf{R}_2)^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^2\mathbf{R}_1\,{}^1\mathbf{p} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10.960155 \\ 10.960155 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{10.960155+10.96015}{\sqrt{2}} \\ \frac{-10.960155+10.9601}{\sqrt{2}} \\ 0 \end{bmatrix}
$$

This matches ${}^2\mathbf{p}$ identically. ■

Solution 1.2

Part 1: Trace Calculation

$$\mathrm{tr}(\mathbf{R}) = R_{11} + R_{22} + R_{33} = 0.999848 + 0.999848 + 1.000000$$

Part 2: Rotation Angle ϕ

Using Eq. (1.24):

$$\cos\phi = \frac{\mathrm{tr}(\mathbf{R}) - 1}{2} = \frac{2.999696 - 1}{2} = \frac{1.999696}{2} = 0.999848$$

$$\phi = \arccos(0.999848) = 0.017436\,\mathrm{rad} \approx 0.9990^\circ \approx 1.00^\circ$$

The structural deflection angle of the boom pin is exactly 1.0°.

Part 3: Eigenvalues and Invariant Axis

The characteristic equation is $\det(\mathbf{R} - \lambda\mathbf{I}) = 0$:

$$\det \begin{bmatrix} 0.999848 - \lambda & -0.017452 & 0 \\ 0.017452 & 0.999848 - \lambda & 0 \\ 0 & 0 & 1 - \lambda \end{bmatrix} = (1 - \lambda) \left[(0.999 \right.$$

Roots:

$$\lambda_1 = 1$$

$$\lambda_{2,3} = 0.999848 \pm j0.017452 = \cos(1^\circ) \pm j \sin(1^\circ) = e^{\pm j(1^\circ)}$$

For $\lambda_1 = 1$:

$$(\mathbf{R} - 1 \cdot \mathbf{I})\mathbf{v}_1 = \begin{bmatrix} -0.000152 & -0.017452 & 0 \\ 0.017452 & -0.000152 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

From rows 1 and 2, $v_x = 0$ and $v_y = 0$, while v_z is arbitrary. Normalizing yields the unit eigenvector:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

The invariant axis of rotation is the spatial Z-axis. ■

Solution 1.3

Part 1: Proof of Positive-Definiteness ($\mathbf{D} \succ 0$)

The matrix is diagonal:

$$\mathbf{D}(d_4) = \begin{bmatrix} D_{11} & 0 \\ 0 & D_{22} \end{bmatrix} = \begin{bmatrix} I_{zz} + M_d(d_4 + c)^2 & 0 \\ 0 & M_d \end{bmatrix}$$

Apply Sylvester's Criterion:

1. First leading principal minor:

$$\Delta_1 = D_{11} = I_{zz} + M_d(d_4 + c)^2$$

Physical constants: $I_{zz} = 287,900\,\mathrm{kg \cdot m^2} > 0$, $M_d = 54,300\,\mathrm{kg} > 0$.

Since $(d_4 + c)^2 \geq 0$ for all real d_4:

$$D_{11} \geq I_{zz} = 287,900 > 0 \quad \forall\, d_4 \in \mathbb{R}$$

1. Second leading principal minor:

$$\Delta_2 = \det(\mathbf{D}) = D_{11} D_{22} = \left[I_{zz} + M_d (d_4 + c)^2 \right] M_d$$

Since $D_{11} > 0$ and $M_d = 54,300 > 0$, $\Delta_2 > 0$ everywhere.

Both leading principal minors are strictly positive for all $d_4 \in (-\infty, \infty)$. Therefore, $\mathbf{D}(d_4) \succ 0$ unconditionally. ■

Part 2: Numerical Eigenvalues across the Stroke

Because $\mathbf{D}$ is diagonal, its eigenvalues are simply its diagonal entries:

$$\lambda_1(d_4) = I_{zz} + M_d (d_4 - 1.32)^2, \quad \lambda_2 = M_d = 54,300\,\text{kg}$$

- At full retraction ($d_4 = 6.75\,\text{m}$):

$$r = 6.75 - 1.32 = 5.43\,\text{m}$$

$$\lambda_1(6.75\,\mathrm{m}) = 287,900 + 54,300(5.43)^2 = 287,900 + 54,300(2$$

$$\lambda_2 = 54,300\,\mathrm{kg}$$

- At full extension ($d_4 = 10.75\,\mathrm{m}$):

$$r = 10.75 - 1.32 = 9.43\,\mathrm{m}$$

$$\lambda_1(10.75\,\mathrm{m}) = 287,900 + 54,300(9.43)^2 = 287,900 + 54,300($$

$$\lambda_2 = 54,300\,\mathrm{kg}$$

Part 3: Ratio of Maximum to Minimum Pitch Inertia

$$\text{Inertia Ratio} = \frac{\lambda_1(10.75\,\mathrm{m})}{\lambda_1(6.75\,\mathrm{m})} = \frac{5,116,522}{1,888,930} \approx 2.7087$$

Control Engineering Significance: The effective pitch inertia of the shovel dipper varies by **271%** ($2.71\times$) across its mechanical crowd stroke. A linear PID controller designed with fixed gains for the retracted position will become under-damped or unstable when fully extended, while a controller tuned for full extension will be sluggish when retracted. This proves why model-based adaptive control and state-dependent gain scheduling are mandatory. ■

Solution 1.4

Part 1: Total Acceleration Derivation

The sensor position in the world frame is:

$$^0\mathbf{p}(t) = \mathbf{R}(t)\,{}^3\mathbf{r}$$

Since ${}^3\mathbf{r} = [s, 0, 0]^T$ is fixed in the handle frame, ${}^3\dot{\mathbf{r}} = \mathbf{0}$ and ${}^3\ddot{\mathbf{r}} = \mathbf{0}$.

Differentiating to find velocity:

$$^0\mathbf{v} = \dot{\mathbf{R}}\,{}^3\mathbf{r} = \mathbf{S}(\boldsymbol{\omega})\mathbf{R}\,{}^3\mathbf{r} = \boldsymbol{\omega} \times {}^0\mathbf{p}$$

Differentiating a second time to find acceleration:

$$^0\mathbf{a} = \frac{d}{dt}\left(\dot{\mathbf{R}}\,^3\mathbf{r}\right) = \ddot{\mathbf{R}}\,^3\mathbf{r}$$

Since $\dot{\mathbf{R}} = \mathbf{S}(\boldsymbol{\omega})\mathbf{R}$, apply the product rule to $\ddot{\mathbf{R}}$:

$$\ddot{\mathbf{R}} = \frac{d}{dt}[\mathbf{S}(\boldsymbol{\omega})\mathbf{R}] = \mathbf{S}(\dot{\boldsymbol{\omega}})\mathbf{R} + \mathbf{S}(\boldsymbol{\omega})\dot{\mathbf{R}} = \mathbf{S}(\dot{\boldsymbol{\omega}})\mathbf{R} + \mathbf{S}(\boldsymbol{\omega})\mathbf{S}(\boldsymbol{\omega})\mathbf{R}$$

Multiplying by $^3\mathbf{r}$:

$$^0\mathbf{a} = \mathbf{S}(\dot{\boldsymbol{\omega}})\mathbf{R}\,^3\mathbf{r} + \mathbf{S}(\boldsymbol{\omega})\mathbf{S}(\boldsymbol{\omega})\mathbf{R}\,^3\mathbf{r} = \dot{\boldsymbol{\omega}} \times {}^0\mathbf{p} + \boldsymbol{\omega} \times (\boldsymbol{\omega}$$

Part 2: Component Separation

For 2D planar rotation where $\boldsymbol{\omega} = [0, 0, \dot{q}_3]^T$, $\dot{\boldsymbol{\omega}} = [0, 0, \ddot{q}_3]^T$, and $^3\mathbf{r} = [s, 0, 0]^T$:

In handle-fixed coordinates (Frame 3):

$$^3\mathbf{a} = \mathbf{R}^T\,^0\mathbf{a} = \mathbf{R}^T\left[\mathbf{S}(\dot{\boldsymbol{\omega}})\mathbf{R}\,^3\mathbf{r} + \mathbf{S}(\boldsymbol{\omega})^2\mathbf{R}\,^3\mathbf{r}\right]$$

Using matrix multiplication:

$$\mathbf{S}(\dot{\boldsymbol{\omega}})\,{}^3\mathbf{r} = \begin{bmatrix} 0 & -\ddot{q}_3 & 0 \\ \ddot{q}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} s \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ s\ddot{q}_3 \\ 0 \end{bmatrix} \quad (\text{Tangential Accele}$$

$$\mathbf{S}(\boldsymbol{\omega})^2\,{}^3\mathbf{r} = \begin{bmatrix} 0 & -\dot{q}_3 & 0 \\ \dot{q}_3 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ s\dot{q}_3 \\ 0 \end{bmatrix} = \begin{bmatrix} -s\dot{q}_3^2 \\ 0 \\ 0 \end{bmatrix} \quad (\text{Centripetal A}$$

Combining both components in Frame 3:

$$ {}^3\mathbf{a} = \begin{bmatrix} -s\dot{q}_3^2 \\ s\ddot{q}_3 \\ 0 \end{bmatrix} $$

- **Tangential Acceleration ($a_y = s\ddot{q}_3$):** Acts perpendicular to the handle, driven by angular acceleration.
- **Centripetal Acceleration ($a_x = -s\dot{q}_3^2$):** Acts along the handle directed inward toward the saddle pivot O_3.
- A radial accelerometer mounted along $+X_3$ will measure exactly $a_x = -s\dot{q}_3^2$ (plus the projection of gravity $g\sin q_3$). ■

Solution 1.5

Part 1: Total Derivative of $D_{11}(d_4)$

Using the multivariable chain rule where d_4 depends on time t:

$$\frac{d}{dt}[D_{11}(d_4)] = \frac{\partial D_{11}}{\partial d_4}\frac{dd_4}{dt} = \frac{\partial}{\partial d_4}\left[I_{zz} + M_d(d_4 + c)^2\right]\dot{d}_4$$

Differentiating with respect to d_4:

$$\frac{\partial D_{11}}{\partial d_4} = 0 + 2M_d(d_4 + c) \cdot (1) = 2M_d(d_4 + c)$$

Therefore:

$$\dot{D}_{11} = 2M_d(d_4 + c)\dot{d}_4 \qquad (1.62)$$

Part 2: Analytical Cancellation Proof

We are given $C_{11} = M_d(d_4 + c)\dot{d}_4$. Compute the difference:

$$\dot{D}_{11} - 2C_{11} = 2M_d(d_4 + c)\dot{d}_4 - 2\left[M_d(d_4 + c)\dot{d}_4\right] = 2M_d(d_4 -$$

The difference vanishes identically to zero. ■

Part 3: Physical Guarantee Against Artificial Energy Drift

In Lagrangian mechanics, the total mechanical energy of the unforced system is $E = T + V = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{D}(q)\dot{\mathbf{q}} + V(q)$. Differentiating total energy with respect to time:

$$\dot{E} = \dot{\mathbf{q}}^T \mathbf{D}\ddot{\mathbf{q}} + \frac{1}{2}\dot{\mathbf{q}}^T \dot{\mathbf{D}}\dot{\mathbf{q}} + \left(\frac{\partial V}{\partial \mathbf{q}}\right)\dot{\mathbf{q}}$$

Substituting the equations of motion $\mathbf{D}\ddot{\mathbf{q}} = \boldsymbol{\tau} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G}$:

$$\dot{E} = \dot{\mathbf{q}}^T \left[\boldsymbol{\tau} - \mathbf{C}\dot{\mathbf{q}} - \mathbf{G}\right] + \frac{1}{2}\dot{\mathbf{q}}^T \dot{\mathbf{D}}\dot{\mathbf{q}} + \mathbf{G}^T\dot{\mathbf{q}} = \dot{\mathbf{q}}^T\boldsymbol{\tau} + \frac{1}{2}\dot{\mathbf{q}}^T \left[\dot{\mathbf{D}} -$$

Because $\dot{\mathbf{D}} - 2\mathbf{C}$ is skew-symmetric (its quadratic form $\dot{\mathbf{q}}^T[\dot{\mathbf{D}} - 2\mathbf{C}]\dot{\mathbf{q}} \equiv 0$ is zero for all $\dot{\mathbf{q}}$):

$$\dot{E} = \dot{\mathbf{q}}^T\boldsymbol{\tau}$$

The Rate of Change of Energy Equals Actuator Power. If $\dot{D}_{11} - 2C_{11}$ did not cancel, the simulated digital twin would either generate numerical energy spontaneously (causing the simulation to blow up to infinity) or dissipate

artificial damping energy (falsely predicting machine stalling). This algebraic cancellation is the mathematical backbone of passivity-based digital twins. ■

1.15 Chapter Summary & Master Revision Checklist

Master Formula Summary Table

Mathematical Concept	Formal Equation	Primary Engineering Use in Digital Twin
Direction Cosine Matrix	$^{A}\mathbf{v} = {}^{A}\mathbf{R}_{B}\,{}^{B}\mathbf{v}, \quad {}^{A}\mathbf{R}_{B} \in SO(3)$	Transforming vectors between world, boom, saddle, and handle frames.
Orthogonality Property	$\mathbf{R}^{-1} = \mathbf{R}^{T}, \quad \det(\mathbf{R}) = +1$	Inverting coordinate systems via matrix transpose without numerical inversion error.
Rotation Axis & Angle	$\mathrm{tr}(\mathbf{R}) = 1 + 2\cos\phi, \quad \mathbf{R}\mathbf{n} = \mathbf{n}$	Extracting structural deflection angles and invariant hinge rotation axes.
Positive-Definite Metric	$\mathbf{D}(q) = \mathbf{D}^{T}(q) \succ 0 \implies \dot{\mathbf{q}}^{T}\mathbf{D}\dot{\mathbf{q}} > 0$	Guaranteeing that physical kinetic energy is positive and forward dynamics can be inverted.
**Time Derivative of	$\dot{\mathbf{R}} = \mathbf{S}(\boldsymbol{\omega})\mathbf{R} = [\boldsymbol{\omega}\times]\mathbf{R}$	Propagating spatial velocities

Mathematical Concept	Formal Equation	Primary Engineering Use in Digital Twin
$\mathbf{R}^{**}$		and kinematic cross-couplings.
Quadratic Energy Rate	$\frac{d}{dt}\left(\mathbf{x}^T\mathbf{A}\mathbf{x}\right) = 2\mathbf{x}^T\mathbf{A}\dot{\mathbf{x}} + \mathbf{x}^T\dot{\mathbf{A}}\mathbf{x}$	Computing power flow and formulating Lyapunov stability functions for observers.

Master Revision Checklist

- [x] Can you define an orthonormal basis and write the formal conditions for $SO(3)$?
- [x] Can you explain why $\mathbf{R}^{-1} = \mathbf{R}^T$ using column vector inner products?
- [x] Can you define the five reference frames of an electric rope shovel ($\mathcal{F}_0$ to $\mathcal{F}_4$) without coordinate ambiguity?
- [x] Do you understand why the determinant of a valid rotation matrix must be $+1$ rather than -1?
- [x] Can you prove why the shovel mass matrix $\mathbf{D}(q)$ must be positive-definite for all physical configurations?
- [x] Can you derive $\dot{\mathbf{R}} = \mathbf{S}(\boldsymbol{\omega})\mathbf{R}$ starting from the identity $\mathbf{R}\mathbf{R}^T = \mathbf{I}$?
- [x] Can you convert a 3×1 angular velocity vector into a 3×3 skew-symmetric matrix and vice versa?
- [x] Can you explain the physical reason why handle pitch inertia varies by over 270% across the crowd stroke?

- [x] Can you prove why $\dot{D}_{11} - 2C_{11} \equiv 0$ prevents artificial numerical energy creation in digital twin solvers?